

Hints for problems

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Only read this hints if you are stuck and want
some extra inspiration

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Problem 1: Entropy

Problem 1A (Discrete case)

The Shannon entropy can be written as

$$H(p) = -\left(s \log s + p \log p + (1 - s - p) \log(1 - s - p)\right).$$

To find its extrema, compute $H'(p) = 0$ and $H''(p)$.